

ACCESS-OM3 panan v2 notes

28/4/26

Some potentially helpful links:

- Repository with notebooks used to generate input files and evaluate model
<https://github.com/claireyung/mom6-panAn-iceshelf-tools>
- Wiki with instructions on how I made the config (quite a few steps)
<https://github.com/claireyung/mom6-panAn-iceshelf-tools/wiki/ACCESS%E2%80%90OM3-config-with-ice-shelves-%E2%80%90-input-files-and-config-details>
- IAF OM3 panan ice shelf config that I ran https://github.com/claireyung/access-om3-configs/tree/ice_shelf_panan-iaf_v2026_2
- How-to-run instructions for that config (running for the first month is a bit different)
https://github.com/claireyung/access-om3-configs/blob/ice_shelf_panan-iaf_v2026_2/ice_shelf_instructions.md
- List of issues to be considered for om3 ice shelf model development
<https://github.com/claireyung/mom6-panAn-iceshelf-tools/issues/49>
 - Discussion about frazil <https://github.com/claireyung/mom6-panAn-iceshelf-tools/issues/43>
- ACCESS-NRI progress towards alpha ice shelf config (basically my old RYF config with a couple of bug fixes, <https://github.com/ACCESS-NRI/access-om3-configs/pull/1078>, Helen/Dougie/Chris were working on this but it's currently blocked by a crash seemingly related to version of JRA forcing)
- ACCESS-OM3 executable used uses this MOM6 version <https://github.com/ACCESS-NRI/MOM6/tree/ice-shelf-dev> and is compiled using spack prerelease tool here <https://github.com/ACCESS-NRI/ACCESS-OM3/pull/142> Requires code changes to nuopc mom cap and ice shelf code described here <https://github.com/ACCESS-NRI/MOM6/issues/29> and here <https://github.com/ACCESS-NRI/MOM6/pull/38>

Issues to prioritise:

- Frazil (?)
 - Before frazil was not treated properly under ice shelves, it just warmed up the water and didn't take the heat out from anywhere (since no sea ice model). Alex from GFDL added it to code in Nov last year after we told them about it, which puts heat flux from warming up supercooled water into a negative melt rate at the ice shelf.
 - I tried this and had basically no refreezing anymore - there was more refreezing with direct freezing in melt param and some supercooled water (frazil off in cavities), which didn't really make sense to me. At OSM MOM6 ice shelf meeting, Bob also thought this was odd. Olga said she had refreezing associated with frazil

in 1/4 degree config, though maybe not in the right places. Alex said that frazil was being passed to the sea ice model despite the sea ice model not meant to have been on under ice shelves. Could be a similar issue with cice. I have not tried his solution yet <https://github.com/mom-ocean/MOM6/commit/45b6d22a2258bc8776d251e93bf0dea207f61ce4>

Alternatively, it could be that the nonlocal refreezing of frazil is important and that by not allowing supercooled water to move, we don't have any refreezing with frazil on.

- At OSM, Xylar said frazil is on in MPAS-Ocean and they get lots of effective refreezing with that, Andrew S said they just turned frazil off in MITgcm, Kaitlin said frazil was modified so that only a little got taken out in each grid cell in open ocean to allow the supercooled water to go further than the first grid cell, though I can't remember if she said it was totally off in cavities. Also MITgcm just does frazil in top layer I think.
- Iceberg flux spreading (?)
 - Anton has PR to spread according to NEMO lagrangian iceberg field, I believe it is not yet implemented in ACCESS-OM3-25km
 - iceberg flux currently has the weird JRA55 box size, affects sea ice formation with that boxy shape, noting that iceberg flux comes with big negative latent heat flux, frazil formation, sea ice formation and brine rejection....
<https://github.com/claireyung/mom6-panAn-iceshelf-tools/issues/49>
- Amundsen-Bellingshausen Cold Bias
 - Big drift over ~20-30 years. Starts okay, current IAF run is better than before with RYF (polynya outside Getz ice shelf) or with WOA initial conditions extrapolated and smoothed (now using Yamazaki conditions extrapolated and not smoothed).
 - Ruth/californian MITgcm people had issues with cold Amundsen depending on initial conditions. Kaitlin also had a cold Amundsen bias in FESOM.
 - Mixing under ice shelf? BL too thin?? Not really sure. ePBL is used, appears to know about ice shelf friction velocity, unsure whether it's actually a sensible mixing parameterisation there <https://github.com/claireyung/mom6-panAn-iceshelf-tools/issues/45#issuecomment-3420519133>
 - Currently, drag coeff is 0.007 (obs suggest 0.001 to 0.01), other panan models use ~0.002 to ~0.006 (Galton-Fenzi et al. 2025)
 - Sampling of oceanic forcing is top 1.08m and distribution of melt fluxes also top 1.08m. These need to be the same value for consistency (especially with frazil off). Also affects open ocean, hence why 1.08m (open ocean top cell) was chosen. But we have many layers within that top 1m due to sigma shelf zstar vertical coordinate
 - Using Holland & Jenkins 1999 melt parameterisation with stability parameter
- Thick sea ice
 - B grid fix Andrew K suggested should work to eliminate ~20km thick sea ice in embayed cells <https://github.com/claireyung/mom6-panAn-iceshelf-tools/issues/49>
 - Need to apply to both sea ice mask and ocean mask and ice sheet input file, and regenerate nuopc mesh files

- Salt restoring (?)
 - currently only on north of the 1000m isobath. have not explored piston velocity modifications
- Really big error logs
 - Pretty annoying, 30GB access-om3.out files. Means control directory needs to not be in home directory since that has a 10GB limit, I store it in gdata instead. Probably slows the model down a bit too writing all that stuff (though it was a similar speed before when I didn't have this error), it seems to have been introduced when I started using this friction velocity bug fix <https://github.com/ACCESS-NRI/access-om3-configs/pull/1078#issuecomment-3912429202>. It complains about making mass but still runs so I've ignored it and deleted error logs afterwards.

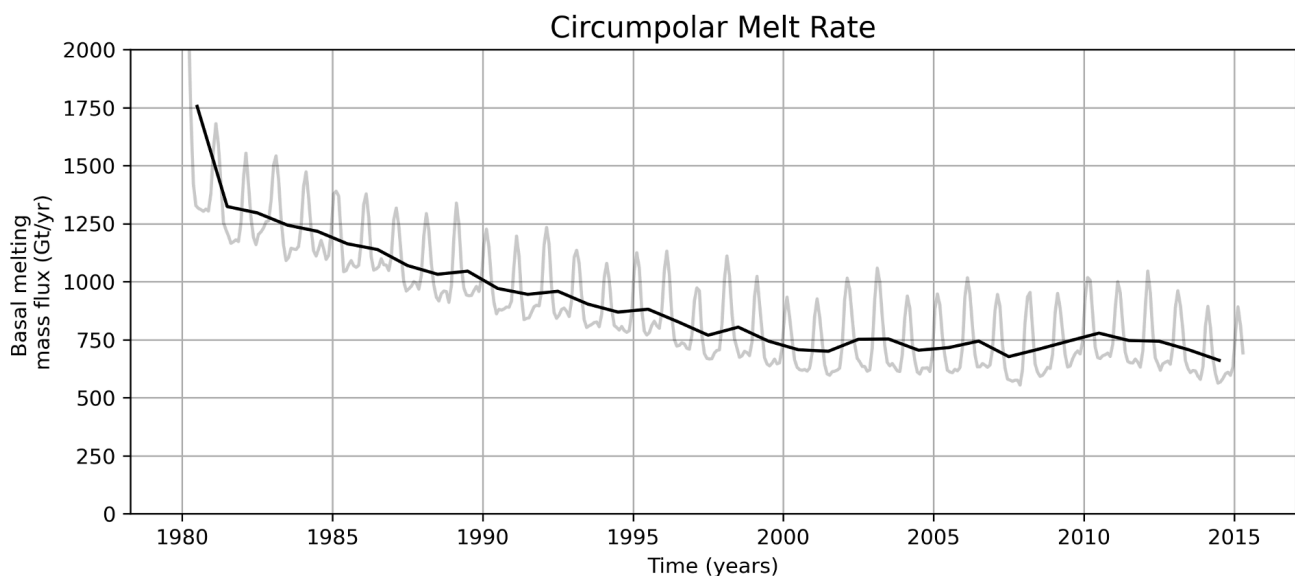
_'''

```
WARNING from PE 0: MOM_diabatic_aux.F90, applyBoundaryFluxesInOut():
Mass created. x,y,dh= -1.609E+02 -8.535E+01 1.316E-12
WARNING from PE 0: MOM_diabatic_aux.F90, applyBoundaryFluxesInOut(): 2
groundings remaining
```

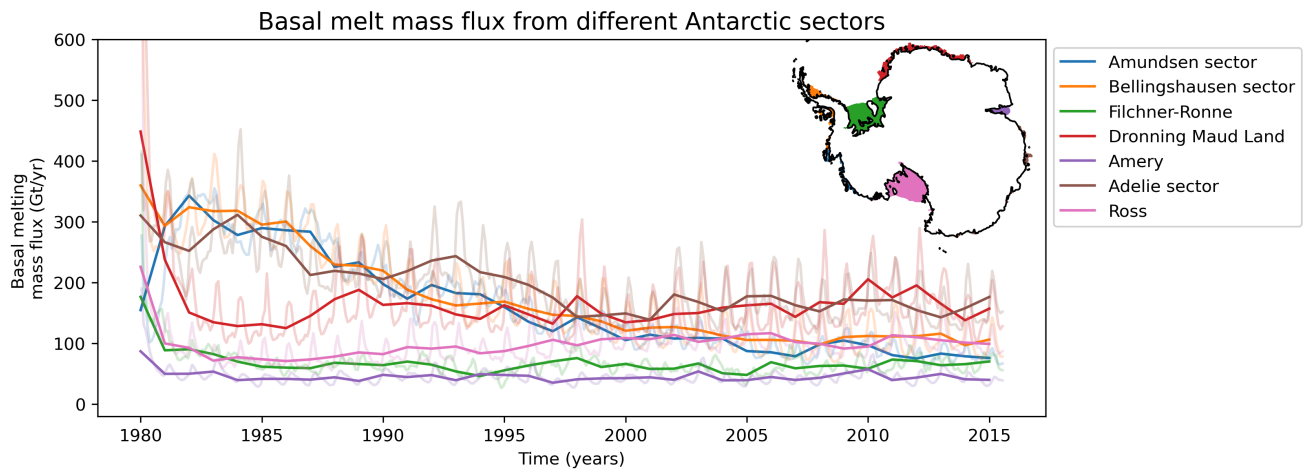
- Potentially these parameters could help: `RESCALE_STRONG_DRAG = True` and `BT_STRONG_DRAG = True` <https://github.com/NOAA-GFDL/MOM6/issues/971#issuecomment-3321662751>

some relevant figures (if not already on github, linked above)

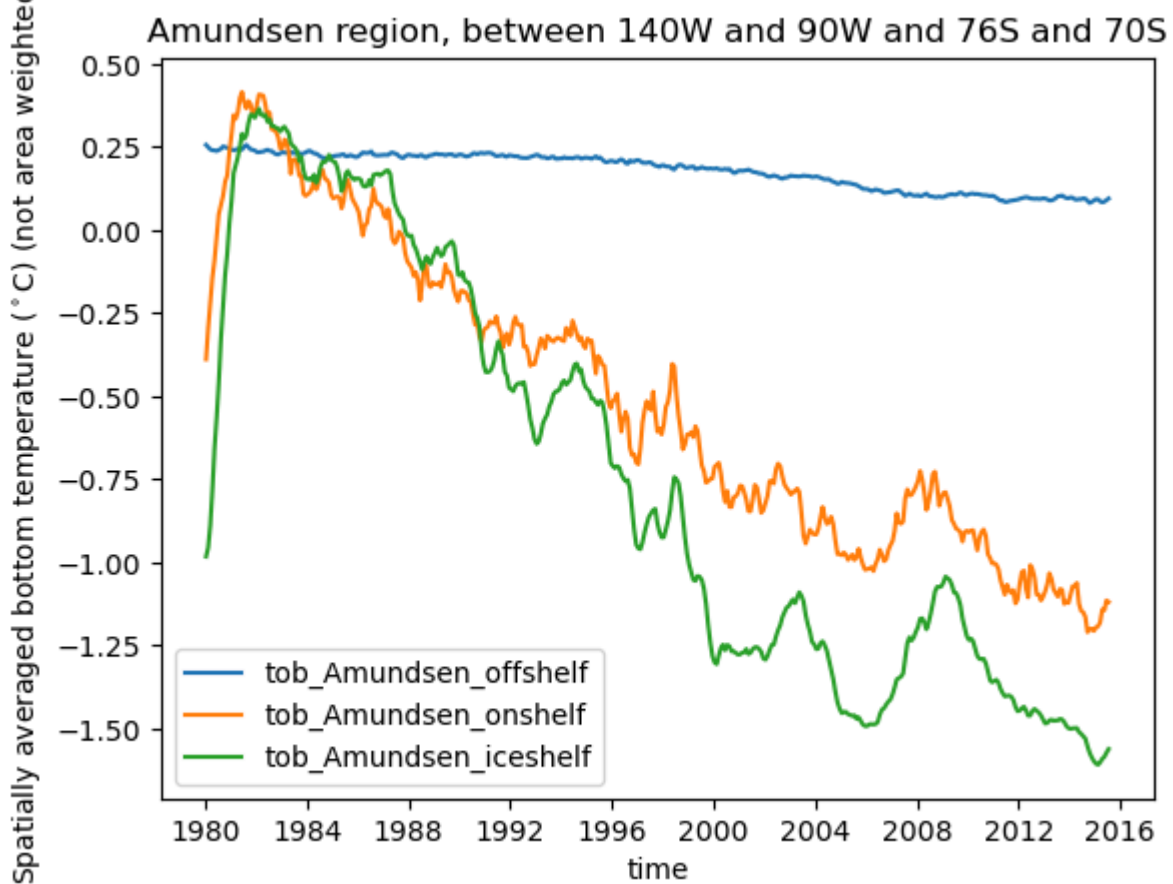
Model melt rate drift in IAF (stabilises by ~2000)



Amundsen/Bellingshausen melt rate drift (blue and orange)

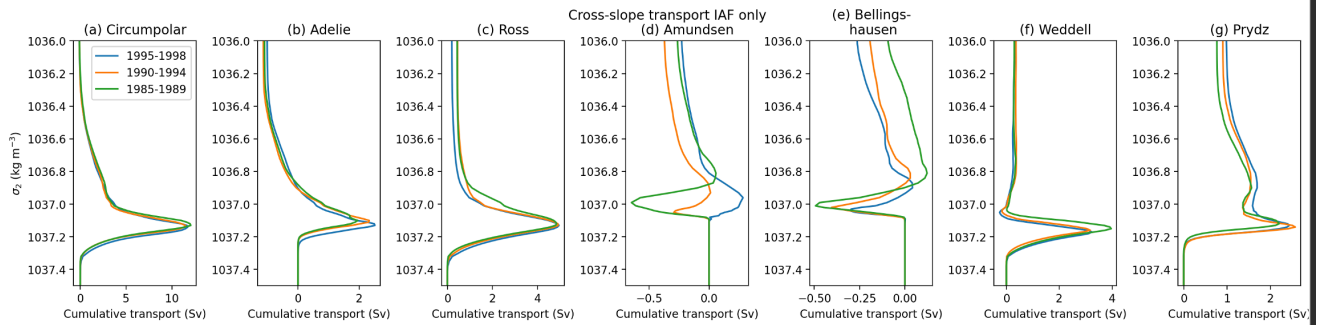


Amundsen bottom temperature drift (off the shelf stays warm-ish though small drift, on the shelf and in the cavities it's rapidly cooling)

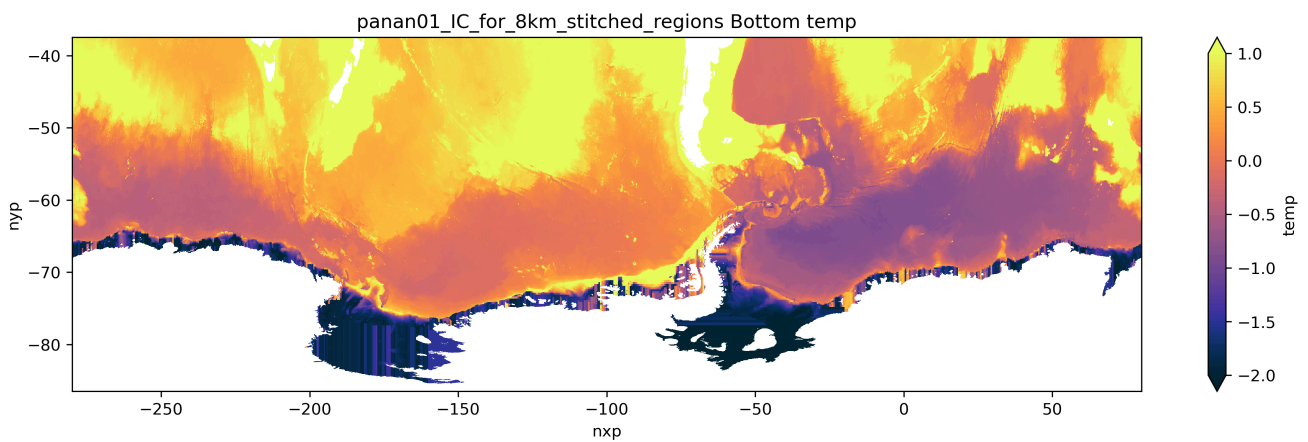


Amundsen volume transport not really showing CDW transport onto shelf after some time (panel d, looked good in the 80s, not so good since 1995 (blue))

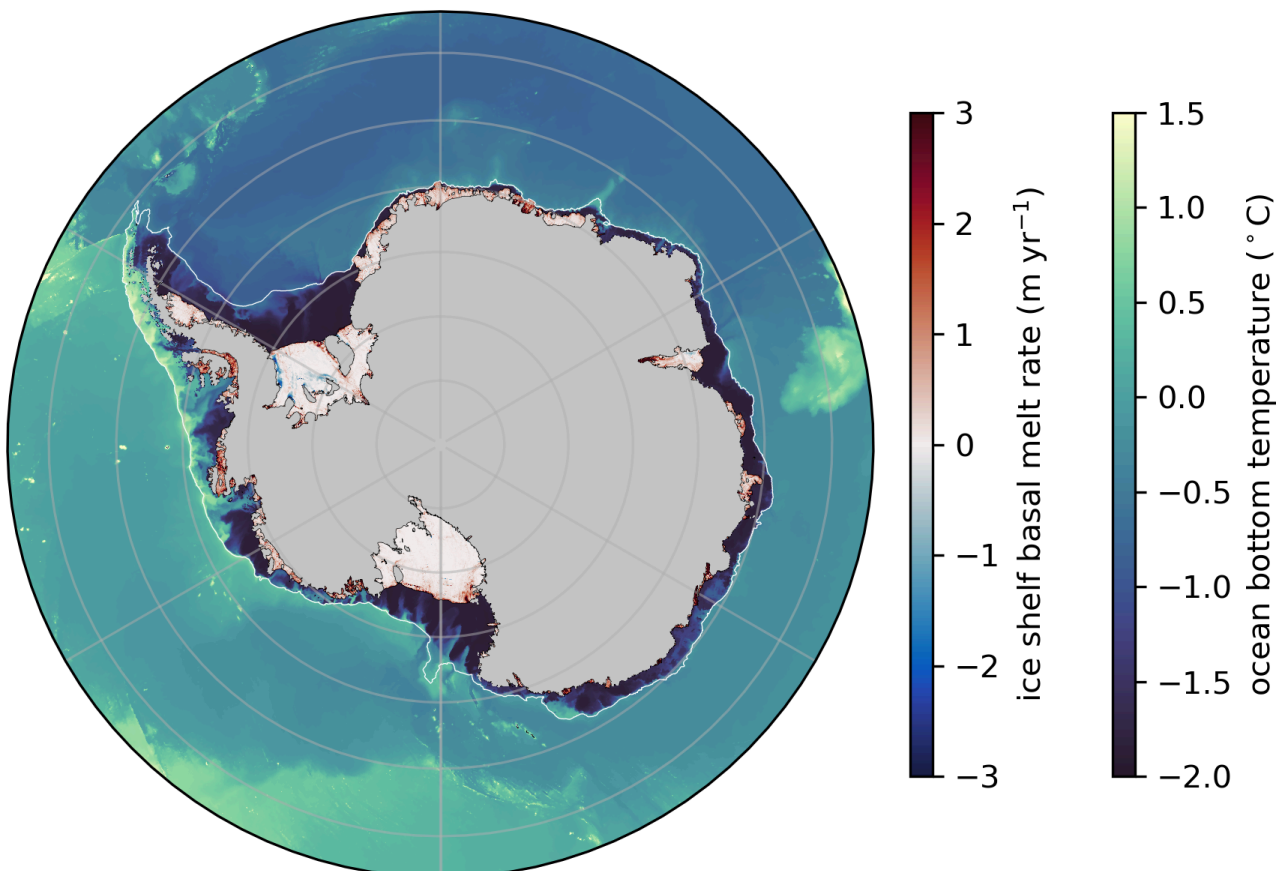
Control IAF in different time periods



Initial conditions in cavities, stripes due to interpolation (changed the direction of interpolation FRIS, Ross and Larsen C) https://github.com/claireyung/mom6-panAn-iceshelf-tools/blob/claire_working/initial-conditions/Yamazaki_IC_into_8km_grid-Copy1.ipynb

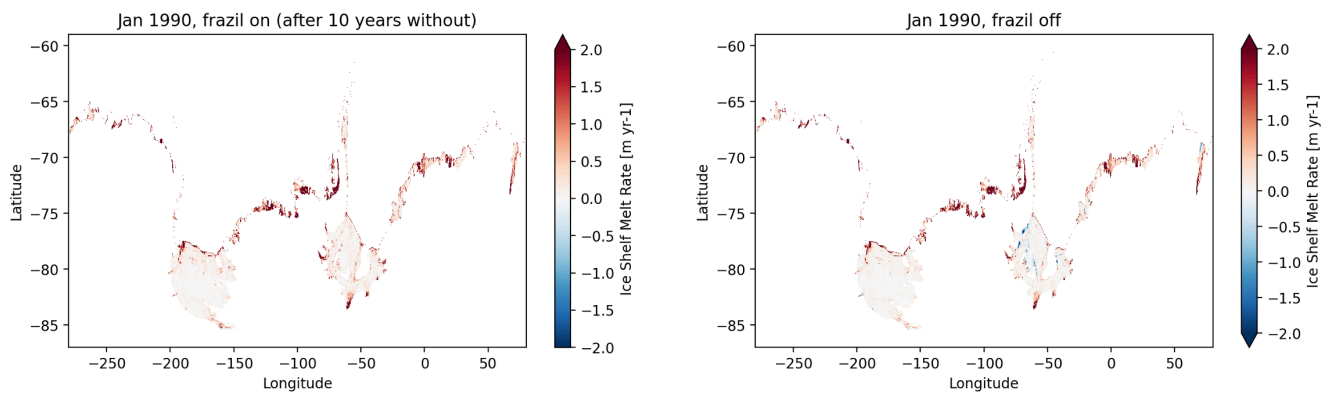


Bottom temperature and melt rate map, mean of 2000-2004

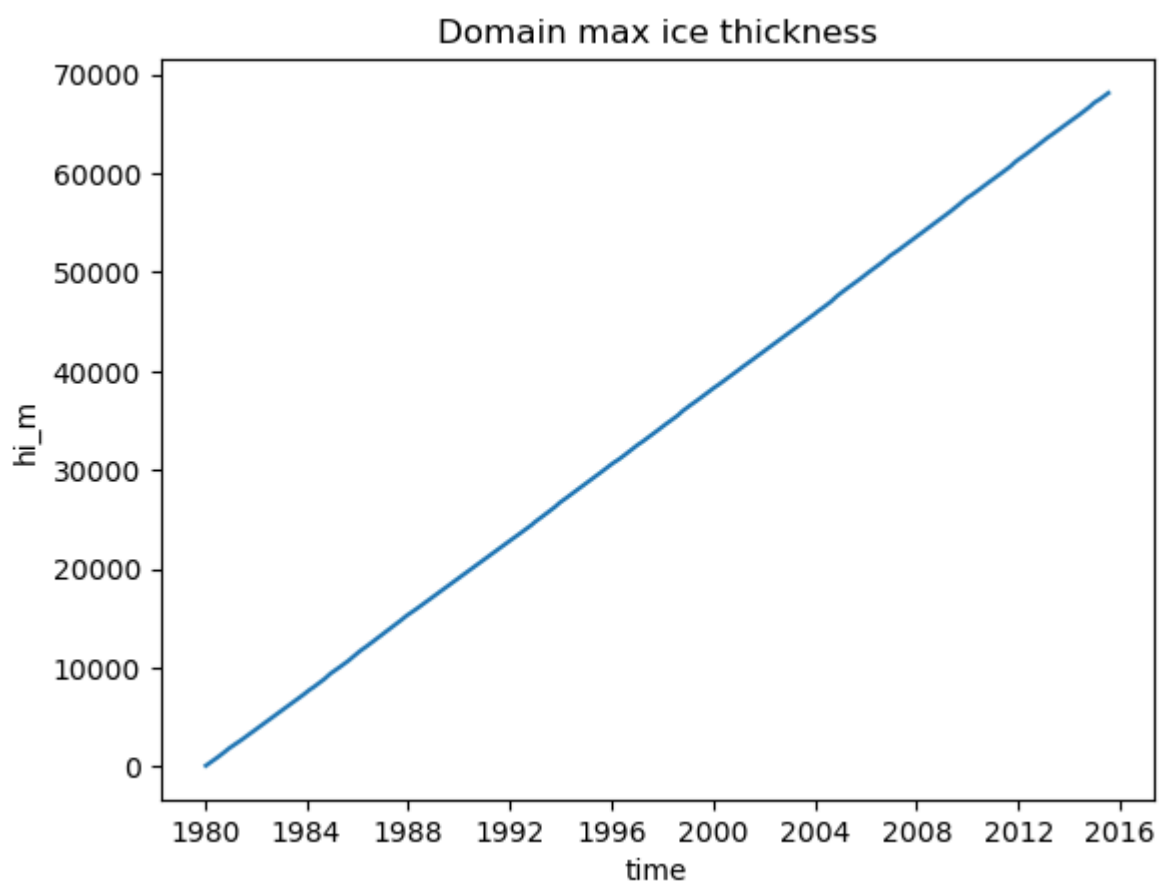


Frazil tests (blue bits all gone on left! weird!)

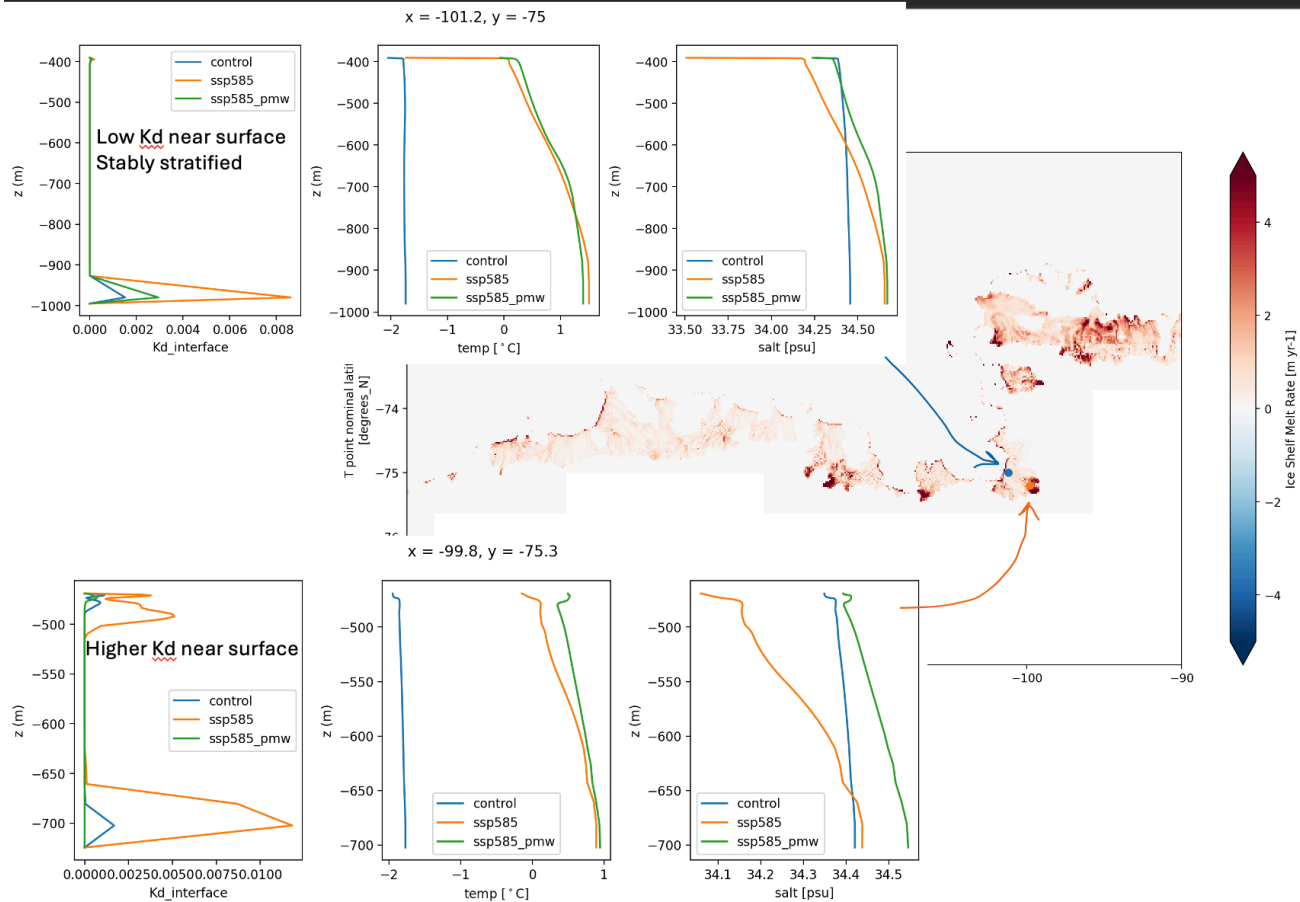
Frazil



Max sea ice thickness timeseries (B grid cice issue)



KD in cavities (PIG example) - i have no idea what it's meant to be



Old tuning experiments with different C_{drag} values and RYF as a guide to how c_{drag} might tune melt rates (tds means thermal driving sampling distance was modified, don't recommend this!! The top two have $c_{drag} = 0.0025$ and 0.003 respectively, new ICs means Yamazaki ICs rather than WOA/ACCESS-OM2 which I originally used and was even colder in Amundsen)

